

Property Settings for Different Controls

Controls	Property	Settings
Form	Caption BackColor StartUpposition	Volt-ampere characteristics As shown Centre of the screen
ADO (Adodc1)	ConnectionString	As explained in the previous VB articles based on MS Access
MSChart	AllowSelections AllowSeriesSelection Boder style ChartType DataSource TitleText	False False VtBorderStyleFixedSingle VtChChartType2DXY Adodc1 VI – Plot
DataGrid	AllowAddNew Allowdelete DataSource TabAction	True True Adodc1 DbgGridNavigation
Shape	Style BorderColor BorderWidth FillColor	Rectangle As shown 3 As shown
Command (inside rectangle shape as shown)	Name Caption Style BackColor	OKButton OK Graphical As shown

A frame control and two Option buttons (inside the frame using copy-and-paste method)

(i) Option button 1	Caption BackColor, ForeColor	Resistor VI As shown
(ii) Option button 2	Caption BackColor, ForeColor	Diode VI As shown

Different controls are combined to draw circuit and associated components and meters

Resistor

(i) Frame	Name BorderStyle BackColor Caption BackStyle BorderWidth FillColor Fillstyle	Resistor None Yellow Blank Opaque 2 As shown Solid
(ii) Shape	BorderColor BorderWidth	As shown 4
(iii) Line control (4 in number, copy and paste method)	BorderColor BorderWidth	As shown 4

Diode

(i) Frame	Name BorderStyle BackColor Caption	Diode None As shown Blank
(ii) Few line controls	Arranged like diode, as shown	

Voltmeter

(i) Shape control	Style BackColor Caption Font size, ForeColor	Circle As shown V As shown
(ii) Label control		

Ammeter

(i) Shape control	Style BackColor Caption Font size, ForeColor	Circle As shown A As shown
(ii) Label control		

Circuit Connections

(use line controls to complete the circuit diagram)

Instead of using different controls to draw the circuit diagram, you can also use separate picture files for the resistor, diode and circuit diagram.

as 'number.' Set field size property to 'single.' Set indexed property of volts as 'yes' (no duplicates).

Program design

In the form_load event, you are required to make a few MSChart control settings:

```
Private Sub Form_Load()  
On Error Resume Next  
R = Val(Text1.Text)  
Plots(1).Value = True  
Plots(1).Value = False  
Plots(0).Value = True  
MSChart1.Plot.UniformAxis = False  
MSChart1.Plot.Axis(VtChAxisIdX).  
AxisTitle = "Voltage"  
MSChart1.Plot.Axis(VtChAxisIdY).  
AxisTitle = "Current"  
MSChart1.Plot.Axis(VtChAxisIdY).  
AxisTitle.TextLayout.Orientation =  
VtOrientationVertical  
OHMS_Adoadjust  
End Sub
```

The resistor current is calculated and updated in the subroutine ohms_adoadjust():

```
Private Sub OHMS_Adoadjust()  
R = Val(Text1.Text)  
Adodc1.RecordSource = "OHMSLAW"  
Adodc1.Refresh  
While Adodc1.Recordset.EOF = False  
Adodc1.Recordset.Fields(1).Value =  
(Adodc1.Recordset.Fields(0).  
Value / R)  
Adodc1.Recordset.MoveNext  
Wend  
DataGrid1.Refresh  
MSChart1.Refresh  
End Sub
```

The diode current is calculated and updated in the subroutine diode_adoadjust():

```
Private Sub DIODE_Adoadjust()  
Dim IAK As Single, VAK As Single  
Adodc1.RecordSource = "DIODEVI"  
Adodc1.Refresh  
While Adodc1.Recordset.EOF = False  
VAK = Adodc1.Recordset.Fields(0).  
Value  
IAK = 50 * (10 ^ -9) * (Exp(VAK /  
(0.26)) - 1)  
Adodc1.Recordset.Fields(1).Value =  
IAK
```